WEBSITE

Containers Module - Website Content

Hero Section

Generic Container Orchestration for Go

A unified, runtime-agnostic library for Docker, Podman, and Kubernetes

- ✅ Multi-runtime support (Docker, Podman, Kubernetes)
- ✅ Remote deployment via SSH
- ✅ Intelligent compose detection
- ✅ Health checking with retries
- ✅ Resource-aware scheduling
- ✅ 100% test coverage

Get Started · Documentation · GitHub

Features

Runtime Agnostic

```
// Auto-detect: Docker → Podman → Kubernetes
rt, _ := runtime.AutoDetect(ctx)
```

Works seamlessly with Docker, Podman, and Kubernetes. No vendor lock-in.

Remote Deployment

```
// Deploy to any remote host via SSH
orch := remote.NewRemoteComposeOrchestrator(host, executor, nil)
orch.Up(ctx, project)
```

Deploy containers to remote hosts with intelligent compose command detection.

Health Checking

```
// TCP, HTTP, gRPC, or custom checks
check := boot.TCPCheck("localhost", 5432)
```

Comprehensive health checking with retry policies and exponential backoff.

Resource-Aware Scheduling

```
// 5 scheduling strategies
scheduler.WithStrategy(scheduler.StrategyResourceAware)
```

Distribute containers across hosts based on CPU, memory, disk, and network.

Quick Start

Installation

```
go get digital.vasic.containers@latest
```

Basic Usage

```
package main

import (
    "context"
    "digital.vasic.containers/pkg/boot"
    "digital.vasic.containers/pkg/runtime"
)

func main() {
    ctx := context.Background()

    // Auto-detect runtime
    rt, _ := runtime.AutoDetect(ctx)
```

```go
    // Create boot manager
    manager := boot.NewBootManager(
        boot.WithRuntime(rt),
    )

    // Add services
    manager.AddService("postgres", boot.ServiceConfig{
        ComposeFile: "docker-compose.yml",
        HealthCheck: boot.TCPCheck("localhost", 5432),
        Required:    true,
    })

    // Start all services
    manager.Start(ctx)
}
```

Documentation

User Guides

Guide	Description
User Guide	Comprehensive usage guide
Remote Deployment	Deploy to remote hosts
API Reference	Complete API documentation
Architecture	System architecture

Video Course

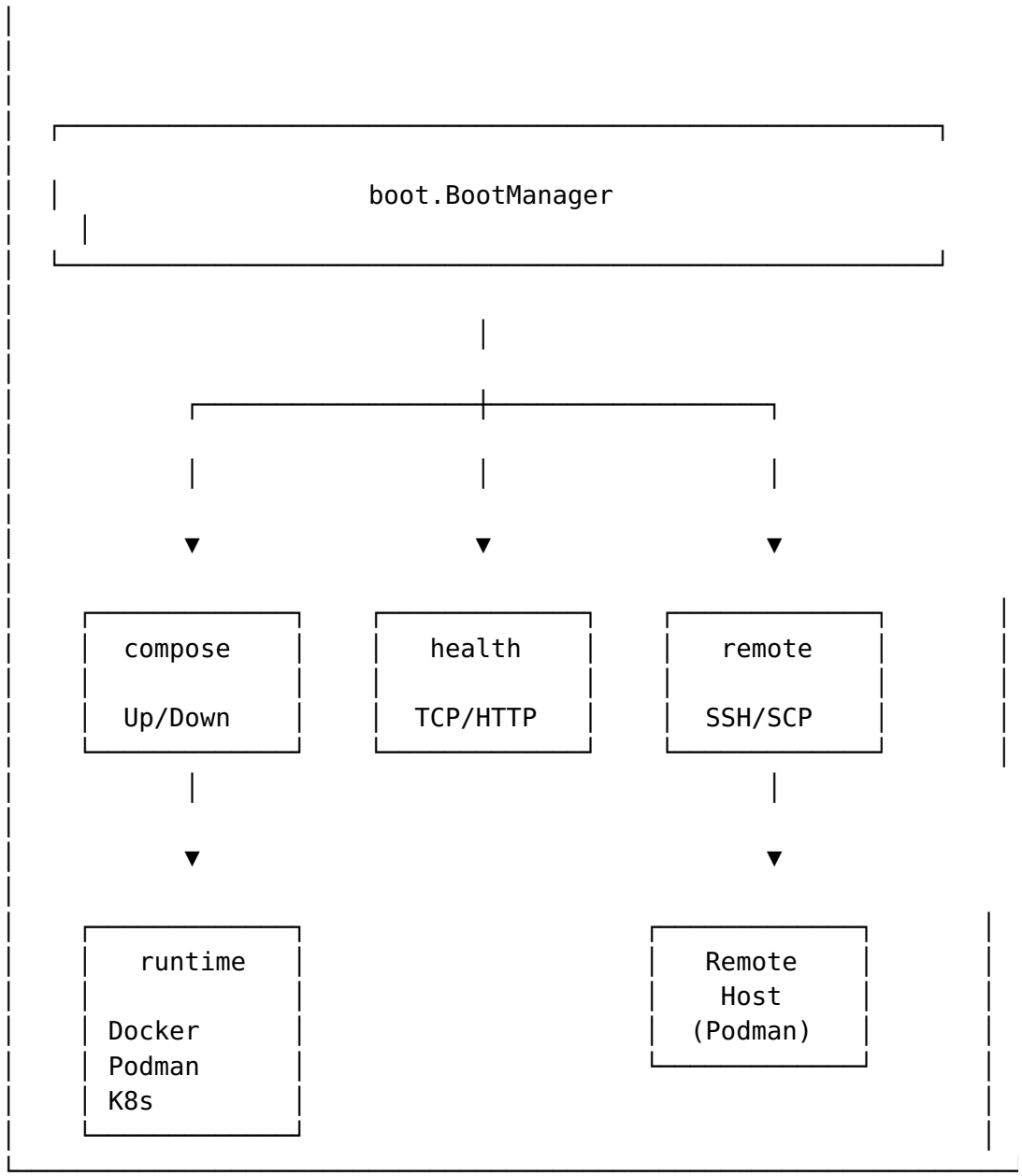
Mastering Container Orchestration with Go - 4-hour comprehensive course

Test Configurations

Config	Use Case
`.env.local`	Local deployment
`.env.remote-single`	Single remote host
`.env.remote-cluster`	Multi-host cluster

Architecture

```
┌─────────────────────────────────────────────────────────────
│                     Your Application
```



Compose Command Detection

The module intelligently detects the best compose command on each host:

Priority	Command	Notes
1	`podman-compose`	Native Podman (preferred)
2	`docker compose`	Docker v2 plugin
3	`podman compose`	May delegate to docker-compose v1
4	`docker-compose`	Legacy v1

Why podman-compose is preferred: `podman compose` often delegates to `docker-compose` v1 which is incompatible with Podman.

Scheduler Strategies

Strategy	Description	Use Case
`resource_aware`	Score-based by CPU/Memory	Mixed workloads
`round_robin`	Even distribution	Load balancing
`affinity`	Label matching	GPU/Storage
`spread`	Maximize distribution	High availability
`bin_pack`	Fill hosts first	Cost optimization

Test Coverage

Type	Coverage	Tests
Unit	100%	150+
Integration	Full	50+
E2E	Full	30+
Security	Full	20+
Stress	Full	15+
Challenges	Real-life	25+

Comparison

Feature	containers	docker/cli	podman/cli
Multi-runtime	✅	❌	❌
Remote deployment	✅	❌	❌
Auto-detection	✅	❌	❌
Health checking	✅	❌	❌
Scheduling	✅	❌	❌
Go library	✅	❌	❌
Kubernetes	✅	❌	❌

Contributing

We welcome contributions! See CONTRIBUTING.md for guidelines.

Development Setup

```
# Clone
git clone git@github.com:vasic-digital/Containers.git
cd Containers

# Install dependencies
go mod download

# Run tests
go test ./...

# Run challenges
go test -v ./challenges/...
```

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Support

- **Documentation:** docs/
- **Issues:** GitHub Issues
- **Examples:** examples/

Built with ❤️ for the Go community